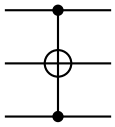
Exercises 5

Exercise 1 (Hadamarding the CNOT gate)**.** *Write down the quantum circuit for the operation* $(H \otimes H)\,\mathrm{CNOT}(H \otimes H)$*. Compute its matrix representation. What gate does this operation amount to?*

Exercise 2 (Funny Toffoli gate)**.** *Write down the explicit matrix for Toffoli gate that is applied in the following way:*



Here the second qubit is the target and the first and third qubits are the controls.

Exercise 3 (Oracles)**.** *Let* $\mathrm{EQ} : \{0,1\}^2 \to \{0,1\}$ *be the equality function on two bits:*

$$\mathrm{EQ}(x_1, x_2) = \begin{cases} 1 & \text{if } x_1 = x_2, \\ 0 & \text{if } x_1 \neq x_2. \end{cases}$$

1. *Write down the matrix for the phase oracle* O_{EQ}*.*
2. *Write down how the bit oracle* U_{EQ} *acts on a general standard basis vector* $|x_1, x_2, y\rangle$ *where* $x_1, x_2, y \in \{0,1\}$*.*
3. *Notice that a bit oracle does not alter the input bits* x_1 *and* x_2 *and that it either flips the output bit* y *or leaves it alone, depending on the value of the function. Using this observation, identify which blocks of the* 8×8 *matrix* U_{EQ} *are non-zero. Then fill in these blocks accordingly.*
4. *Express* EQ *using* XOR *and* NOT*. Using this expression, write* $\mathrm{EQ}(x_1, x_2)$ *as an algebraic expression of* x_1 *and* x_2 *(modulo* 2*).*
5. *Based on the above expression, write down a quantum circuit for* U_{EQ} *that involves only* CNOT *and X gates.*

You can access the following exercises through the course *Canvas* page. There you can also find a link to *CodeGrade* where you should submit your solutions (they will be checked and graded automatically). You should implement your solutions in Cirq. Always make sure you base your solution on the template file provided on *CodeGrade*!

Exercise 4 (`Cirq`: Implementing SWAP)**.**

Exercise 5 (`Cirq`: Deutsch's algorithm with a phase oracle)**.**

Exercise 6 (`Cirq`: Deutsch's algorithm with a binary oracle)**.**

Exercise 7 (`Cirq`: OR-circuit on two qubits)**.**

Exercise 8 (`Cirq`: OR-circuit on three qubits)**.**